

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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Section: CSM-B

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1. TITLE

Design of a Responsive Business Landing Page using HTML and CSS (Flexbox and Media Queries)

2. AIM

To design a responsive single-page business website consisting of a header with navigation, a home banner, a services section with card-based layout, and a contact section, using HTML for structure and CSS (Flexbox and media queries) for styling and responsiveness, without any JavaScript.

3. OBJECTIVE

The objectives of this experiment are:

- To structure a single-page website using semantic sections linked by anchor navigation.
- To style a page consistently using an external CSS file and CSS custom resets.
- To arrange service cards horizontally using Flexbox.
- To apply hover effects on links and buttons using CSS pseudo-classes.
- To make the layout responsive for smaller screens using a CSS media query.

4. THEORY

A landing page is commonly organised as a sequence of full-width sections — a header, a hero/home section, a features or services section, and a contact section — each identified by an id attribute so that navigation links can jump to them using in-page anchors (`href="#id"`). A universal selector reset (`* { margin: 0; padding: 0; box-sizing: border-box; }`) removes inconsistent browser default spacing and makes width calculations predictable. Flexbox, applied to a container with `display: flex`, arranges child elements such as service cards in a row and distributes space between them using properties like `gap` and `justify-content`, without needing floats or manual positioning. Pseudo-classes such as `:hover` let CSS change a link or button's appearance in response to mouse interaction, purely through styling rules. Responsive design is achieved with a media query (`@media (max-width: 700px)`) that overrides earlier rules once the viewport width drops below a threshold, for example stacking the navigation links vertically and changing the card container's `flex-direction` from row to column so cards stack on narrow screens.

5. ALGORITHM / PROCEDURE

1. Create `index.html` and link an external `style.css` file inside the `<head>`, along with a responsive viewport meta tag.
2. Build a `<header>` containing an `<h1>` site title and a `<nav>` with anchor links to Home, Services, and Contact sections.
3. Add a `<section class="home" id="home">` with a heading, an introductory paragraph, and a "Get Started" call-to-action link styled as a button.
4. Add a `<section class="services" id="services">` containing a `<div class="cards">` with three `<div class="card">` blocks, each with a heading, a description, and a "Learn More" link.
5. Add a `<section class="contact" id="contact">` with a heading, a short message, and a "Contact Us" mailto link styled as a button.
6. Add a `<footer>` with a copyright paragraph.
7. In `style.css`, apply a universal box-sizing reset and set a base font family, background colour, and smooth scroll behaviour on the html element.
8. Style the header with a solid background colour, centred text, and spaced-out navigation links, including a `:hover` colour change.
9. Style the home and contact sections with centred text and padding, and style `.btn / .card-btn` with a background colour, padding, rounded corners, and a `:hover` state.
10. Use `display: flex` with `gap` and `justify-content` on `.cards` to lay the three service cards out horizontally, and give each `.card` a fixed percentage width, padding, rounded corners, and a box-shadow.
11. Add a media query for `max-width: 700px` that stacks the navigation links vertically and switches `.cards` to `flex-direction: column` with full-width cards.

12. Open index.html in a browser at desktop width and then narrow the browser (or use device emulation) to verify the responsive behaviour.

6. CODE EXPLANATION

The page is built from a single index.html linked to an external style.css, with no JavaScript. The <header> holds the site title and a <nav> of anchor links that use the browser's native smooth-scrolling (enabled via scroll-behavior: smooth on the html selector) to jump between sections. The .home and .contact sections share centred, padded styling and each contain a call-to-action link styled with the shared .btn class — a solid background, padding, and rounded corners that turn a plain anchor into a button-like element, with a :hover rule that swaps the background colour. The .services section uses a flex container (.cards) to lay three .card divs out in a row with consistent spacing (gap) and centring (justify-content), and each card has its own padding, rounded corners, and box-shadow to appear as a raised panel; a .card-btn link inside each card reuses the same button styling as .btn. A <footer> with a dark background and centred text closes the page. Finally, an @media (max-width: 700px) block overrides the navigation and card layout for narrow viewports: nav becomes a block-level element so links stack vertically, and .cards switches from flex-direction: row to column with each .card widened to 90%, so the three cards stack on top of one another on small screens instead of being squeezed side by side.

7. PROGRAM

```
index.html
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Creative Web</title>

  <!-- CSS File -->
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- Header -->
  <header>
    <h1>Creative Web</h1>

    <nav>
      <a href="#home">Home</a>
      <a href="#services">Services</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <!-- Home Section -->
  <section class="home" id="home">

    <h2>Welcome to My Website</h2>

    <p>
      We create simple and beautiful websites.
    </p>

    <a href="#services" class="btn">Get Started</a>

  </section>
```

```

<!-- Services Section -->
<section class="services" id="services">

    <h2>Our Services</h2>

    <div class="cards">

        <div class="card">
            <h3>Web Design</h3>

            <p>
                We create attractive and responsive website designs
                for different types of businesses.
            </p>

            <a href="#contact" class="card-btn">Learn More</a>
        </div>

        <div class="card">
            <h3>Development</h3>

            <p>
                We develop fast, simple and user-friendly websites
                using modern web technologies.
            </p>

            <a href="#contact" class="card-btn">Learn More</a>
        </div>

        <div class="card">
            <h3>Support</h3>

            <p>
                We provide website support and maintenance to keep
                your website running smoothly.
            </p>

            <a href="#contact" class="card-btn">Learn More</a>
        </div>

    </div>

</section>

<!-- Contact Section -->
<section class="contact" id="contact">

    <h2>Contact Us</h2>

    <p>
        Have a project in mind? Get in touch with us today.
    </p>

```

```

        <a href="mailto:example@gmail.com" class="btn">
            Contact Us
        </a>

    </section>

    <!-- Footer -->
    <footer>

        <p>
            &copy; 2026 Creative Web. All Rights Reserved.
        </p>

    </footer>

</body>

</html>
style.css
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

html {
    scroll-behavior: smooth;
}

body {
    font-family: Arial, sans-serif;
    background: #f4f4f4;
    color: #222;
}

/* Header */
header {
    background: #2563eb;
    color: white;
    text-align: center;
    padding: 30px;
}

header h1 {
    margin-bottom: 15px;
}

/* Navigation */
nav a {
    color: white;
    text-decoration: none;
    margin: 0 12px;
}

nav a:hover {
    color: red;

```

```

}

/* Home */
.home,
.contact {
    background: white;
    text-align: center;
    padding: 60px 20px;
}

.home h2,
.contact h2 {
    color: #2563eb;
    margin-bottom: 15px;
}

.home p,
.contact p {
    margin-bottom: 25px;
}

/* Buttons */
.btn,
.card-btn {
    background: #2563eb;
    color: white;
    padding: 10px 20px;
    text-decoration: none;
    border-radius: 5px;
    display: inline-block;
}

.btn:hover,
.card-btn:hover {
    background: palevioletred;
}

/* Services */
.services {
    padding: 50px 20px;
    text-align: center;
}

.services h2 {
    margin-bottom: 30px;
}

.cards {
    display: flex;
    gap: 20px;
    justify-content: center;
}

.card {
    background: white;
    padding: 25px;
    width: 30%;

```

```

border-radius: 8px;
box-shadow: 0 3px 8px #ccc;
}

.card h3 {
color: #2563eb;
margin-bottom: 12px;
}

.card p {
margin-bottom: 20px;
}

/* Footer */
footer {
background: #222;
color: white;
text-align: center;
padding: 20px;
}

/* Mobile */
@media (max-width: 700px) {
nav a {
display: block;
margin: 8px;
}

.cards {
flex-direction: column;
align-items: center;
}

.card {
width: 90%;
}
}

```

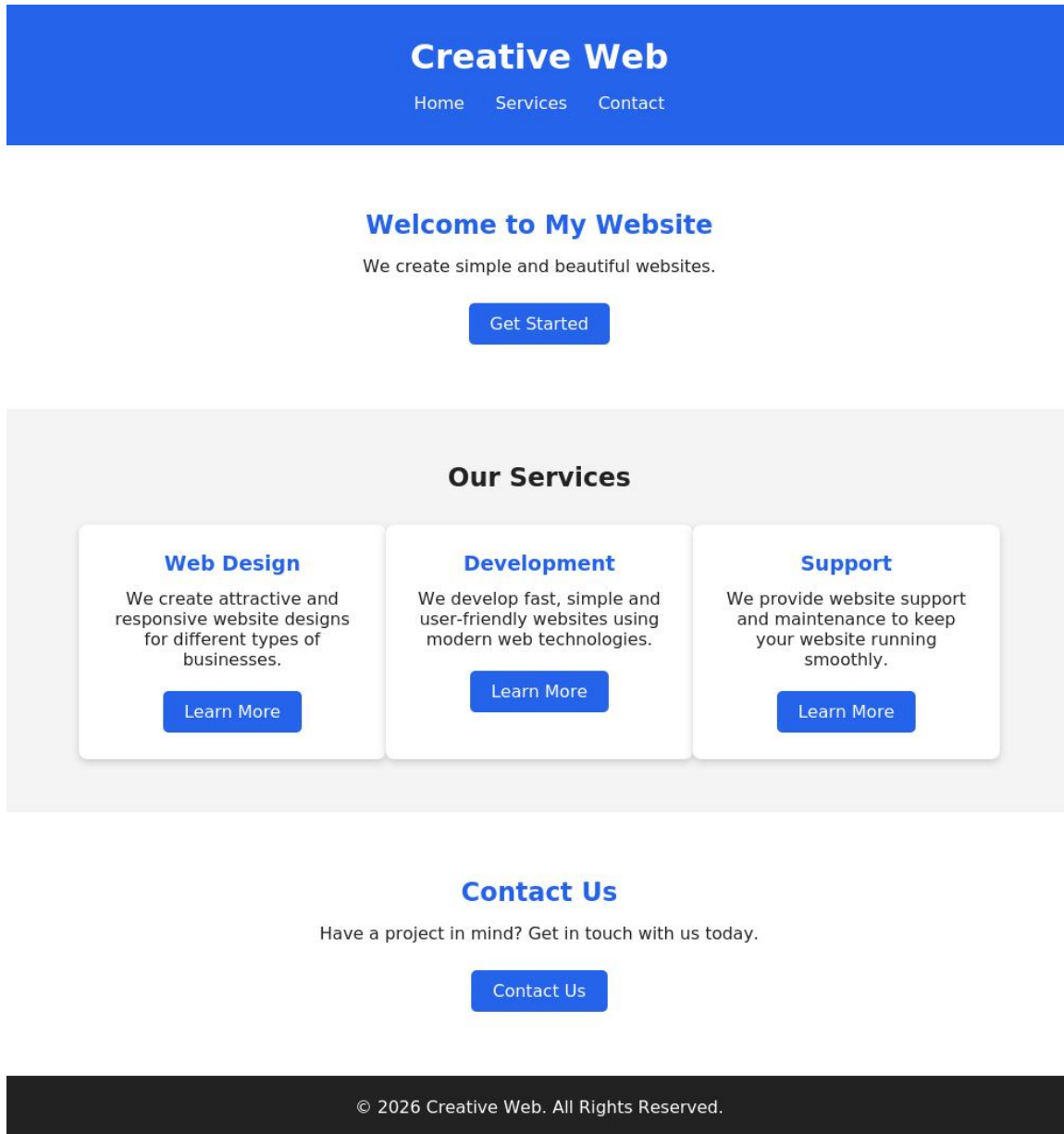
8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open index.html at desktop width	Header, home banner, three service cards in a row, contact section, and footer all display correctly	Page rendered correctly with all sections visible	Pass
TC-02	Click "Home", "Services", or "Contact" in the navbar	Page smoothly scrolls to the corresponding section	Smooth scroll to correct section observed	Pass
TC-03	Hover over a nav link or a .btn / .card-btn button	Nav link text turns red; buttons change background to palevioletred	Hover colour changes applied correctly	Pass
TC-04	Resize the browser window to below 700px width	Nav links stack vertically and service cards switch from a row to a stacked column at 90% width	Responsive layout triggered correctly at the breakpoint	Pass

TC-05	Click the "Contact Us" button	Default mail client opens a new email addressed to example@gmail.com	Mailto link triggered the mail client correctly	Pass
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9. OUTPUT

Desktop view (nav bar, home banner, service cards in a row, contact, and footer):



Mobile / narrow-screen view (stacked navigation and stacked service cards via media query):

10. RESULT

Thus, a responsive single-page business landing page was successfully designed using HTML and CSS. Flexbox was used to lay out the service cards, hover effects were applied to links and buttons, and a media query made the navigation and card layout adapt correctly to narrower screens. The output was verified in a browser at both desktop and mobile widths.